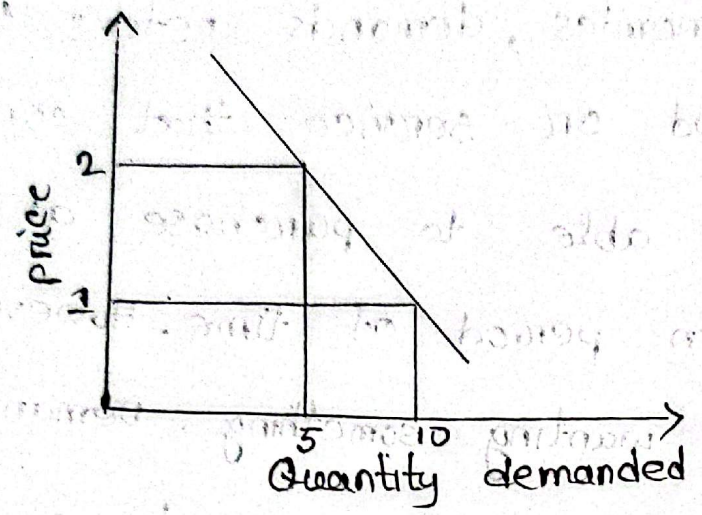
Demand

Demand:- In economics, demands refers to the quantity of a good or service that consumers are willing and able to purchase at various prices over a given period of time. However, demand is more than just wanting something. Demand exists only when a person not only wants a product but is also willing and financially able to pay it. for it.

Law of Demand:- The law of demand states that, all else being equal, there is an inverse relationship between the price of a good or services and the quantity demanded by consumers. As the price of a good rises, the quantity demanded tends to decrease; conversely, as the price falls, the quantity demanded generally increases.

price ↑ → quantity ↓

price ↓ → quantity ↑



The law of demand.

Factors affecting Demands:

1. Consumer income:- Demand for normal goods rises with higher income, while demand for inferior goods may decrease as income increases.
2. Consumers Preference and Tastes:- Changes in tends, fashion, marketing and cultural influences affect consumer preferences, which can increase or decrease demand.
3. Prices of related Goods:- Demand is influenced by the prices of related goods.
   - Complementary good: If the price of a

complementary good decreases, demand for the related good increases because they are used together.

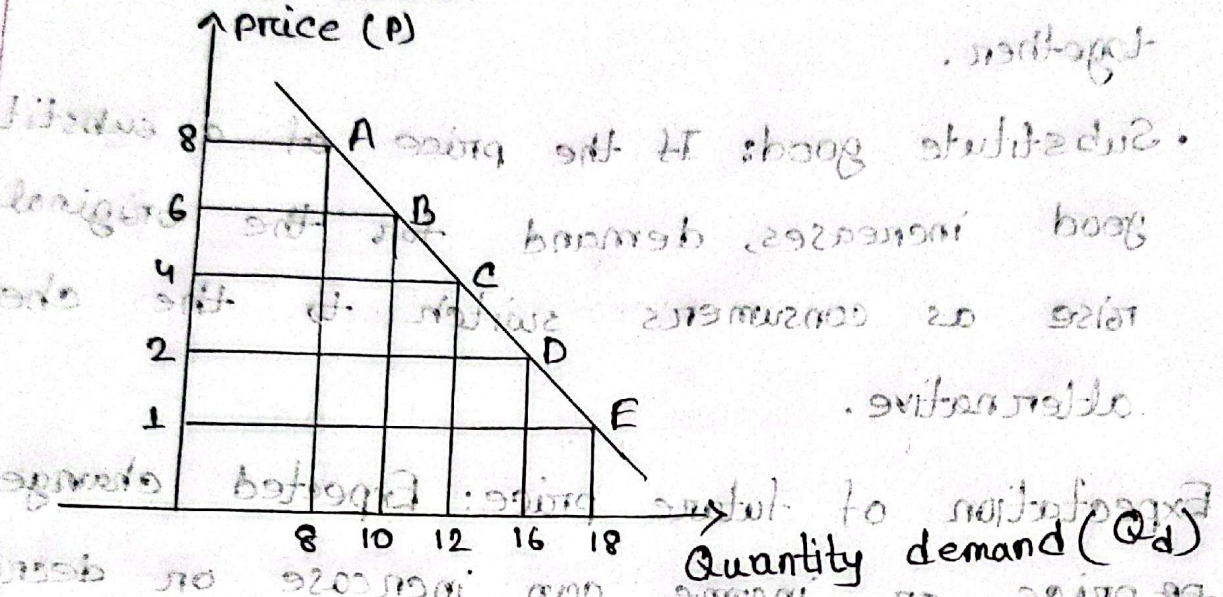
- Substitute good: If the price of a substitute good increases, demand for the original good rise as consumers switch to the cheaper alternative.

4. Expectation of future price: Expected change in price or income can increase or decrease current demand.

5. Seasonal factors :- Weather and season change demand for certain products

Demand Schedule and Demand Curve:

- **Demand curve:**



- **Demand Schedule:**

Point	Price (P)	Quantity Demand (Q_d)
A	8	8
B	6	10
C	4	12
D	2	16
E	1	18

Demand Function: A demand function represents the relationship between the quantity of a good demanded and the factors affecting it, primarily prices.

$$Q_d = a - bp$$

where,

Q_d = Quantity demand

a = intercept

b = slope

p = price

Math: Create a demand function on your own and draw a demand curve and demand schedule from it.

=> Demand function:

$$Q_d = 50 - 5p$$

if $p = 1 \Rightarrow Q_d = 45$

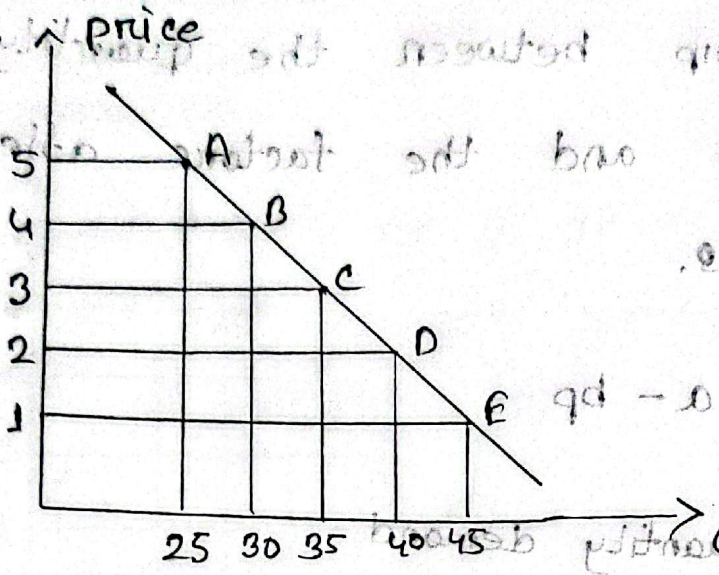
if $p = 2 \Rightarrow Q_d = 40$

if $p = 3 \Rightarrow Q_d = 35$

if $p = 4 \Rightarrow Q_d = 30$

if $p = 5 \Rightarrow Q_d = 25$

Demand curve -

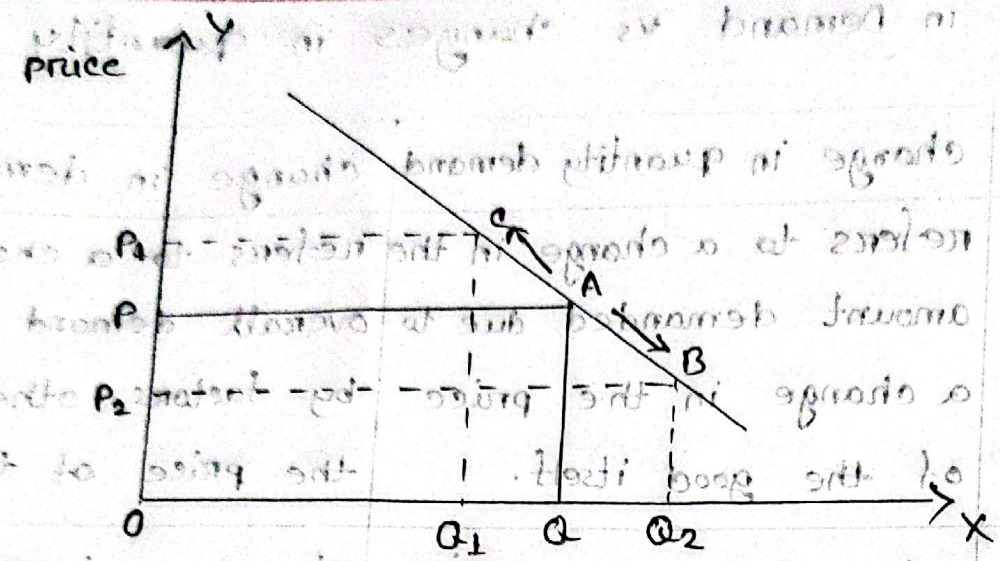


Demand schedule -

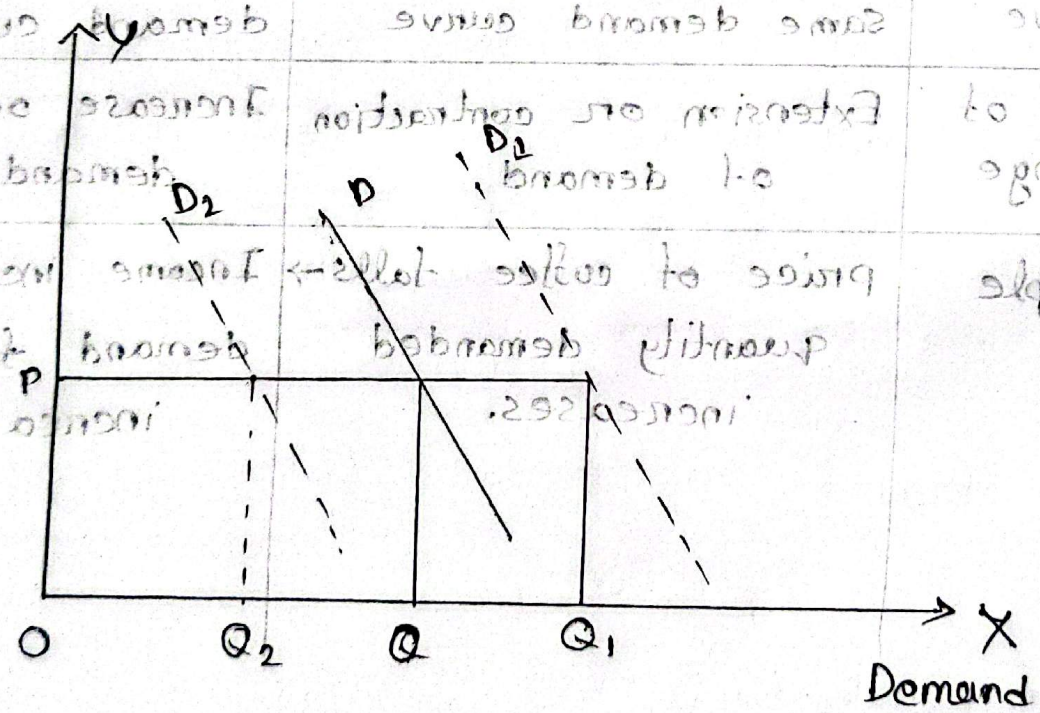
point	P	Q_d
A	5	25
B	4	30
C	3	35
D	2	40
E	1	45

Changes in Demand Vs changes in quantity Demand:

Factors	change in quantity demand	change in demand
Meaning	refers to a change in the amount demanded due to a change in the price of the good itself.	refers to a change in overall demand caused by factors other than the price of the good.
cause	change in own price only.	change in income, tastes, population, expectation, or price of related goods.
Effect on curve	movement along the same demand curve	shift of the entire demand curve.
Types of change	Extension or contraction of demand.	Increase or decrease in demand.
Example	price of coffee falls → quantity demanded increases.	Income increases → demand for coffee increases.



Change in Quantity Demand



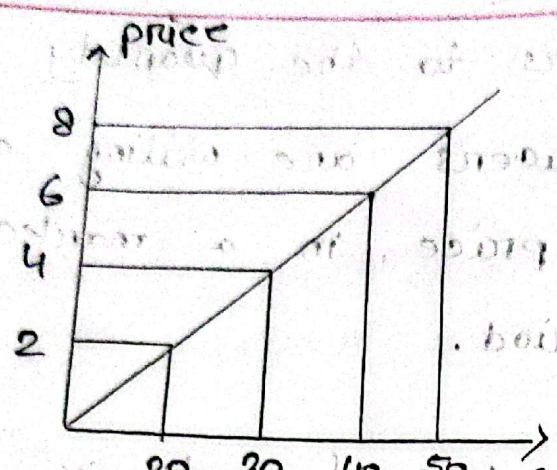
Chang in Demand

Supply

Supply: Supply refers to the quantity of a commodity that producers are willing and able to sell at a given price in a market during a specific time period.

Law of supply: The law of supply is a fundmental principle in economics that describes the relationship between the price of a good or service and the quantity that producers are willing to supply, assuming all other factors remain constant.

It states that, other things being equal, the quantity of good supplied increases as the price of the goods increases, and decreases as the price decreases. So this reflects a direct (positive) relationship between price and quantity supplied.



Factors Affecting Supply

1. Price of the good: The price of the good directly affects the quantity supplied. As price of the good increases, the quantity supplied increases and vice varsa.

2. Cost of the production: The cost of production inputs such as labor, raw materials, and machinery directly affects the supply of a good.

3. Technology: Technological advancement improve production effeciency by reducing production costs and increase outputs.

5. Expectations of future prices: Producer's expections about future prices influence current supply decisions. If prices are expected to rise, supply may decrease now, while expectations of falling prices encourage producers to increase present supply.

6. Number of producers: The number of producers in a market directly affects the total supply of a good. An increase in producers raises overall supply, while a decrease in producers reduces the market supply.

Supply Function : The supply function in economics is a mathematical expression that represents the relationship between the quantity of a good or service supplied and the various factors affecting it.

$$Q_s = a + bp$$

where,

a = intercept

b = slope

p = price

Math: (market equilibrium)

Demand function : $Q_d = 50 - 5P$

Supply function : $Q_s = 10 + 3P$

Find out Equilibrium price and output.

$\Rightarrow$

$$Q_d = Q_s$$

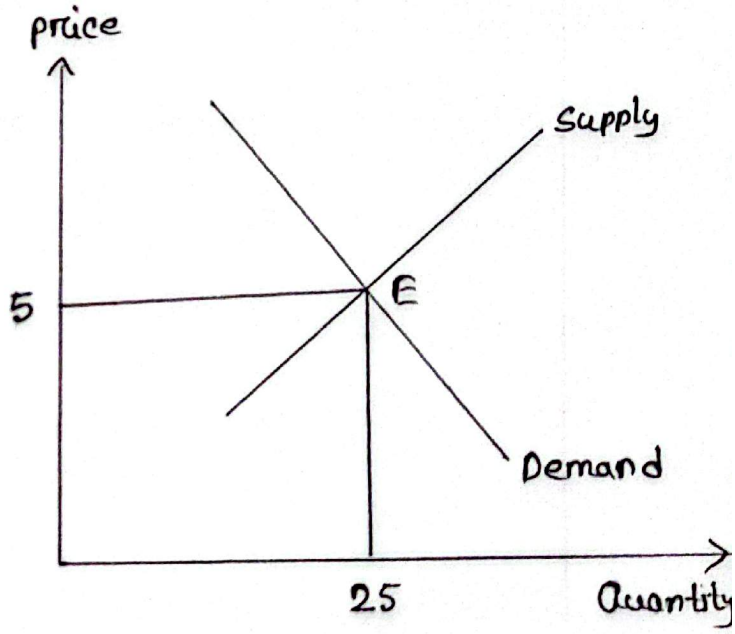
$$\Rightarrow 50 - 5P = 10 + 3P$$

$$\Rightarrow 8P = 40$$

$$\Rightarrow P = 5$$

$$\therefore Q_d = 50 - 5 \times 5$$
$$= 25$$

$$\therefore Q_s = 10 + 3 \times 5$$
$$= 25$$



$\therefore$ The equilibrium quantity, $Q_e = 25$

Since $Q_d = Q_s$, the solution is correct

Elasticity

Definition: Elasticity is a measure of how much buyers and sellers respond to changes in market conditions. It allows us to analyze supply and demand with greater precision.

$$\text{Elasticity} = \frac{\%\,\Delta Q_d}{\%\,\Delta P}$$

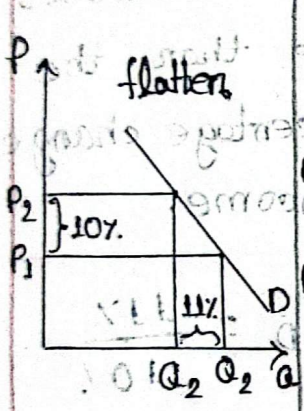
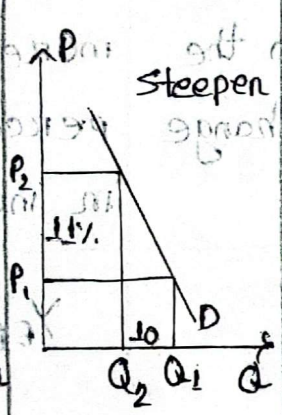
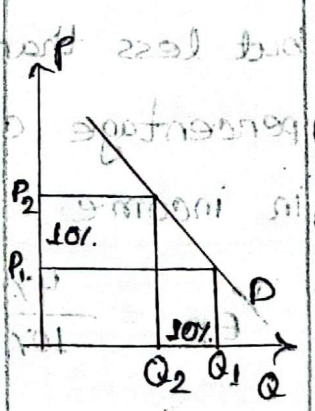
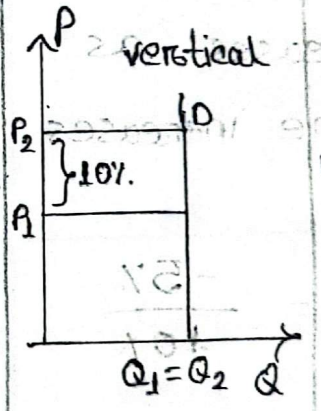
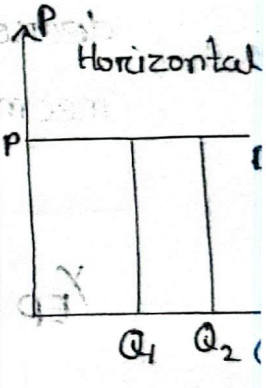
$\%\,\Delta Q_d$ = Percentage change in Quantity Demanded.

$\%\,\Delta Q_d$ = Percentage change in price

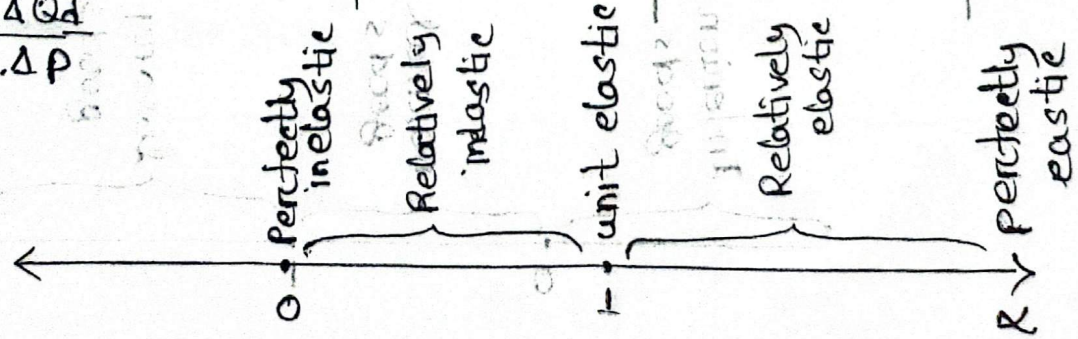
There are three types of elasticity of demand:

- Price elasticity.
- Income elasticity.
- Cross elasticity.

Price Elasticity: It defined as the responsiveness of quantity demanded to a change in price.

Relatively elastic	Relatively inelasic	Unit elastic	Perfectly inelastic	Perfectly elastic
Quantity demanded changes more than price changes.	Q_d changes exactly proportionally to price chages	Q_d changes less than price changes	Q_d does not change at all, no matter the price	Even a tiny price change causes an infinite change in Q_d
				
$PED = \frac{11\%}{10\%}$ $PED > 1$	$PED = \frac{10\%}{11\%}$ $PED < 1$	$PED = \frac{10\%}{10\%}$ $PED = 1$	$PED = \frac{0}{10\%}$ $PED = 0$	$PED = \frac{10\%}{0}$ $PED = \infty$

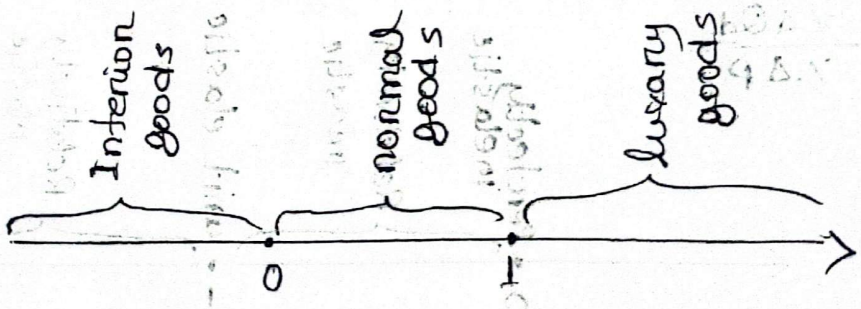
$$PED = \frac{\%\Delta Q_d}{\%\Delta P}$$



Income Elasticity: Income elasticity of demand measures how much the quantity demanded of a good responds to a change in consumers income.

$Y_{ED} = \frac{\%\Delta Q_d}{\%\Delta Y}$ → percentage change in Income

Inferion Goods	Normal Goods	Luxary goods
Demand decreases as income increases	Demand increases but less than the percentage change in income	Demand increases more than the percentage change in income.
$Y_{ED} = \frac{-5\%}{10\%}$ $Y_{ED} < 0$	$Y_{ED} = \frac{4\%}{10\%}$ $0 < Y_{ED} < 1$	$Y_{ED} = \frac{11\%}{10\%}$ $Y_{ED} > 1$



Cross Elasticity: Cross elasticity of demand measures how the quantity demanded of one good changes in response to a change in the price of another related good. It indicats whether the two goods are substitutes, complements or independent.

Complementary goods

B	A
tea	Sugar
P↑	Q_d↓

Substitutes goods

B	A
Coffee	Tea
P↑	Q_d↑

$$X_{ED} = \frac{\%\Delta Q_d^A}{\%\Delta P^B}$$

Q_d^A = Quantity demand of good A

P^B = price of good B

S.C | W.C | Ig | W.S | S.S

-1 | 0 | 1

Strong Substitude	Weak substitude	Independent good	Weak complement	Strong Complement
A small price change in one good causes a large demand change in the other	A price change in one good causes a small demand change in the other	No relationship between the two goods	A price change in one good slightly impacts demand for the other	A price change in one good significantly impact demand for the other
$X_{ED} = \frac{12\%}{10\%}$	$X_{ED} = \frac{8\%}{10\%}$	$X_{ED} = \frac{0}{10\%}$	$X_{ED} = \frac{-5\%}{10\%}$	$X_{ED} = \frac{-15\%}{10\%}$
$X_{ED} > 1$	$0 < X_{ED} < 1$	$X_{ED} = 0$	$-1 < X_{ED} < 0$	$X_{ED} < -1$

Math (price elasticity)

Q1 When the price of CD increased from $20 to $22, the quantity of CDs demanded decreased from 100 to 87. What is the price elasticity of demand for CDs?

$$\Rightarrow \%\Delta P = \frac{22-20}{20} * 100 = \left[\frac{P_2 - P_1}{P_1} \times 100\right]$$

$P_1 = 20$, $P_2 = 22$, $Q_1 = 100$, $Q_2 = 87$

$$\approx 10\%$$

$$\%\Delta Q_d = \frac{87-100}{100} * 100 = \left[\frac{Q_2 - Q_1}{Q_1} * 100\right]$$

$$= -13\%$$

$$\therefore PED = \left|\frac{10\%}{-13\%}\right| \left|\frac{-13\%}{10\%}\right|$$

$$\approx 1.3\%$$

When PED > 1 that is Relatively elastic.

Math (Income Elasticity)

- Consumer's income increase from $2000 to $2500.
- The quantity of a good demanded rise from 100 units to 130 units.

$$\Rightarrow \%\Delta Q_d = \frac{Q_2 - Q_1}{Q_1} \times 100 \quad \left| \begin{array}{l} Q_1 = 100 \\ Q_2 = 130 \end{array} \right.$$

$$= \frac{130 - 100}{100} \times 100$$

$$= 30\%$$

$$\%\Delta Y = \frac{Y_2 - Y_1}{Y_1} \times 100 \quad \left| \begin{array}{l} Y_1 = 2000 \\ Y_2 = 2500 \end{array} \right.$$

$$= \frac{2500 - 200}{2000} \times 100$$

$$= 25\%.01 \times \frac{100 - 100}{100}$$

$$\therefore Y_{ED} = \frac{\%\Delta Q_d}{\%\Delta Y}$$

$$= \frac{30\%}{25\%}$$

$$= 1.2$$

Here $Y_{ED} > 1$, this good is classified as a luxury good.

Math (Cross elasticity)

- Good A : coffee
- Good B : Tea(~~rreate~~ related goo)

1. Initial values:
   - Initial quantity of coffee demanded $(Q_1^A) = 100$ cup
   - Initial price of tea $(P_1^B) = \$2$

2. New values:
   - New quantity of coffee demanded $(Q_2^B) = 120$ cup
   - New price of tea $(P_2^B) = \$3$

$$\Rightarrow \%\Delta Q_d^A = \frac{Q_2^A - Q_1^A}{Q_1^A} * 100$$

$$= \frac{120 - 100}{100} \times 100$$

$$= 20\%$$

$$\%\Delta P^B = \frac{P_2^B - P_1^B}{P_1^B} * 100$$

$$= \frac{\$3 - \$2}{2} \times 100$$

$$= 50\%$$

$$\therefore X_{ED} = \frac{\%\Delta Q_d^A}{\%\Delta P^B}$$

$$= \frac{20\%}{50\%}$$

$$= 0.4$$

Here $0 < X_{ED} <$

So, 0.4 means coffee and tea are weak substitutes.

Math (cross elasticity)

- Initial quantity demanded of bread (Q_1^A) = 50 units
- New quantity demanded (Q_2^A) = 55 units
- Initial price of butter (P_1^B) = $4
- New price of butter (P_2^B) = $5

Calculate - 1. $\% \Delta Q_d^A$

2. $\% \Delta P^B$

3. XED

Determine the relationship (substitute complement or independent)

$\Rightarrow$

1. $\% \Delta Q_d^A = \frac{Q_2^A - Q_1^A}{Q_1^A} * 100\%$

$= \frac{55 - 50}{50} * 100$

$= 10\%$

2. $\% \Delta P^B = \frac{P_2^B - P_1^B}{P_1^B} * 100$

$= \frac{5 - 4}{4} * 100$

$= 25\%$

3. $XED = \frac{\% Q_d{}^A}{\% P^B}$

$= \frac{10\%}{25\%}$

$= 0.4$

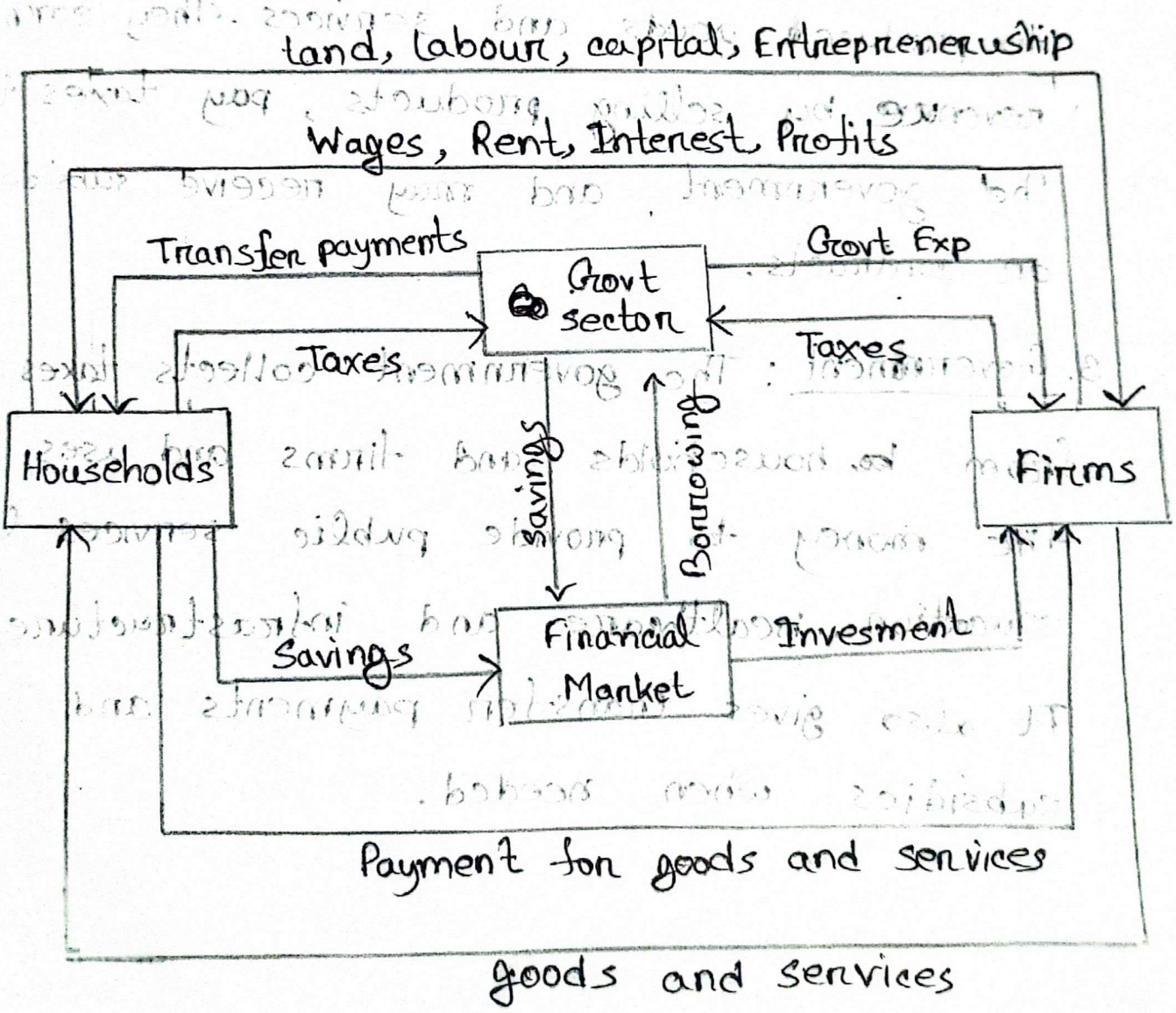
Here $XED > 0$, positive X_{ED}

A positive X_{ED} indicates that the two goods are substitutes, not complements. This means that when the price of one good (butter) increases, the quantity demanded of the other good (bread) increases.

Circular flow of Income

Three sector circular flow of Income:

The three sector circular flow of income model is an expanded form of the two sector model that includes the government sector. It shows how income, goods, services, and factors of production move among households, firms, and the government through taxes, spending and economic activities.



Components:

1. Households - It provide labor, land, capital and entrepreneurship to firms and earn income such as wages, rent, interest and profit. They use this income to buy goods and services and pay taxes to the government.

2. Firms: Firms use resources from households to produced goods and services. They earn revenue by selling products, pay taxes to the government and may receive subsides on contracts.

3. Government: The government collects taxes from households and firms and uses the money to provide public services like education, healthcare, and infrastructure. It also gives transfer payments and subsidies when needed.

Iteractions:

1. Household - Government: Households pay taxes to the government, and in return receive public services and financial support.

2. Firm - Government: Firms pay corporate taxes and may receive subsidies or contracts from the government, which affect their business activities.

The inclusion of the Government sector significantly affects the overall economic situation.

Total Expenditure, $E = C + I + G_r$ ——— ①

C = consumption expenditure

I = investment expenditure

G_r = government expenditure

Total income, $Y = C + S + T$ ——— ②

C = consumption

S = Savings

T = taxes

Since expenditure (E) made must be equal to the income received (Y)

$\therefore \quad C + I + G_r = C + S + T$

$\Rightarrow \quad I + G_r = S + T$

$\Rightarrow \quad G_r - T = S - I$

$T > G_E \Rightarrow$ Budget surplus

$G_r E > T \Rightarrow$ Budget Deficit [G_E = govt expenditure]

Injection: Injections are additions to the circular flow of income that increase money circulation in the economy.

1. Investment (I): Spending on capital goods like machine, buildings, and infrastructure.
2. Government Spending (Gr): Government expenditure on services as education, healthcare and defense.
3. Exports (X): Selling domestic goods and services to foreign countries, bringing income to the economy.

$$\text{Injection} = X + I + Gr$$

Leakages: Leakages are withdrawals from the circular flow of income that reduce money circulation in the economy.

1. Savings(S): Income saved instead of spent on good and services

2. Taxes (T): Payments made to the government that reduce spending power.

3. Imports (M): Spending on foreign goods and services, sending money abroad.

$$\text{Leakage} = S + T + M$$

For economic equilibrium:

$$S + T + M = I + G + X$$

When injections are greater than leakages, the economy grows. When leakages exceed injections, the economy may contract.

GDP

$$gdp = C + I + G + Nx \quad [Nx = X - M]$$

Math

1. Suppose, In an open economy; consumption (C) = 400, Investment (I) = 150, Government Spending (G_t) = 100, Exports (X) = 70, Imports (M) = 50, Taxes (T) = 120, Savings (S) = 150. Find out GDP and verify the equilibrium conditions of injections and leakages.

⇒

$$GDP = C + I + G_t + Nx$$
$$= C + I + G_t + (X - M)$$
$$= 400 + 150 + 100 + (70 - 50)$$
$$= 670$$

$$\text{Leakages} = S + T + M$$
$$= 150 + 120 + 50$$
$$= 320$$

$$\text{Injection} = I + G_t + X$$
$$= 150 + 100 + 70$$
$$= 320$$

Since, Leakages = Injection

The equilibrium condition satisfied.

2. Same question -> values are different.

=> Given,

$C = 4000$, $I = 1800$, $G = 2000$, $X = 700$

$M = 500$, $T = 1200$, $S = 1800$

$$GDP = C + I + G + (X - M)$$
$$= 4000 + 1800 + 2000 + (700 - 500)$$
$$= 8000$$

$$\text{Leakages} = S + T + M$$
$$= 1800 + 1200 + 500$$
$$= 3500$$

$$\text{Injection} = I + G + X$$
$$= 1800 + 2000 + 700$$
$$= 4500$$

$\therefore$ Leakages $\neq$ Injection

The equilibrium conditions not satisfied

chapter-1

Introduction

Economics: Economics is the social science that studies how people, businesses and governments use limited resources to satisfy unlimited wants.

 # Relationship between scarcity and choice:

Scarcity means that resources are limited, but human wants are unlimited. Since people can't have everything they want, resources must be used carefully.

Due to scarcity, individuals, firms, and government must make choices about how to use their limited resources.

Scarcity creates the need for choice. When resources are limited, choosing one thing means giving up another thing. Every choice involves an opportunity cost, which is the ~~best~~ value of the next best alternatives that is sacrified.

Central Problems of an Economy -

The main objective of an economy is to satisfy human wants. But resources are limited compared to unlimited wants. Because of this scarcity, every society must make important decisions about the use of resources. These decisions are known as the central problems of an economy.

The four major central problems are discussed below:

1. What to produce

This problem refers to deciding:

- Which goods and services should be produced
- How much of each should be produced

Since resources are limited, an economy can't produce everything people want. Society must choose between necessary goods and luxury goods. Priority is usually to essential goods needed by people.

CamScanner

2. How to produce:

This problem is about deciding:

- Which method of production should be used
- What combination of resources should be applied.

Generally, there are two main methods of production:

- Labor-intensive method - use more human labor
- Capital-intensive method - use more mechinery and technology.

An economy must choose the method that is most efficient and economical according to its available resources.

3. For Whom to produce

This problem is about deciding:

- How goods and services will be distributed among people

Since, production is limited, not everyone can receive the same amount of goods and services. Distributi manly depends on income, purchasing power, and wealth.

Thus, an economy must decide who will get what share of national output.

4. What provision should be made for economic growth

This problems refers to:

- How to increase the economy's productive capacity in the future.

An economy must decide how many resources should be used for present consumption and future development through investment and savings.

How the central problems of an economy are solved -

⇒ The central problems of an economy are solved through different economic systems. There three main types of ~~econoly economy~~ economics systems are free market economy, comand economy, and mixed economy. Each system uses a different method to decide how resources will be allocated and how goods and services will be produced and distributed.

1. Free market economy

A free market economy is an economic system where economic activities are controlled mainly by supply and demand with little government interference. Firms produce goods and services to earn profit, while prices are determined freely in the market. ~~comple~~ Competition among bussinesses increases efficiency and ~~quanti~~ quality.

2. ~~Demand~~ Command Economy

A command economy is an economic system where the government controls major economic decisions. The government decides what to produce, how much to produce and how resources will be used. The main goal is social and welfare and equality.

3. Mixed economy

A mixed economy is a combination of free market and command economy. In this system, both private businesses and government play important roles in economic activities. Market forces work together with government control and regulations. Most countries in the modern world follow a mixed economic system.